

HAIPENG CAI

TENURED ASSOCIATE PROFESSOR

Department of Computer Science and Engineering (CSE)
School of Engineering and Applied Sciences (SEAS)
University at Buffalo, The State University of New York (UB)

CONTACT INFORMATION

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RESEARCH INTERESTS

My primary research interests are in software/system security and software engineering, with a current focus on program analysis and AI/ML for quality/security-assurance applications to real-world, complex software systems of highly decoupled composition (distributed systems, mobile software platforms/apps, and multi-language software).

PROFESSIONAL EXPERIENCE (Academic)

State University of New York at Buffalo, Buffalo, NY, USA

Tenured Associate Professor in the CSE Department

08/2024 -

- Cross-language program analysis and security
- Security defense for language runtime systems
- Analysis and security of mobile and distributed software systems
- AI for software/system security

Washington State University, Pullman, WA, USA

Huie-Rogers Endowed Chair in Computer Science

07/2023 - 08/2024

Tenured Associate Professor in the School of EECS

08/2022 - 08/2024

- Program analysis and security of highly-decoupled software systems (e.g., multilingual, mobile, distributed software)
- Machine/deep learning for and by software/system security analysis (e.g., data generation, benchmarking, multilingual code representation learning)

Assistant Professor in the School of EECS

08/2016 - 08/2022

- Program analysis of distributed software systems and its applications
- Cross-language code analysis and its applications
- Mobile software security, maintenance, and evolution
- Data augmentation and benchmarking for program vulnerability analysis

Virginia Tech, Blacksburg, VA, USA

Postdoctoral Associate in the Department of Computer Science

08/2015 - 08/2016

Supervisors: Dr. Barbara G. Ryder and Dr. Danfeng (Daphne) Yao

- Dynamic study of Android programming paradigms and application security
- Static inter-application dataflow analysis for collusion-attack defense
- Large-scale, parallel Android inter-application risk detection and ranking

The University of Notre Dame, South Bend, IN, USA

Research Assistant in the Department of Computer Science and Engineering

08/2012 - 08/2015

- Program dependence analysis (abstraction, approximation, prioritization, quantification)
- Change impact prediction in software evolution
- Dynamic impact analysis of common distributed programs

The University of Southern Mississippi, Hattiesburg, MS, USA

Research Assistant in the School of Computing

09/2010 - 06/2012

- Diffusion tensor MRI (DTI) visualization and its dependence of seeding resolution in 3D tractography, effect of display settings on DTI visualization, sculpturing approach to tract-of-interest (TOI) selection and manipulation
- MRI image and DTI data processing, 3D DTI tractography full pipeline maintenance

Zhejiang University, Hangzhou, Zhejiang, China

Research Assistant at the State Key Lab of CAD & Computer Graphics

09/2006 - 07/2008

- Motion tracking in streaming video via Internet, parallel video monitoring and alarming
- Moving shadow detection and texture synthesis for vision-based human-computer interaction
- 3D geometrical modeling for computer aided costume design, surface morphing and expanding

SUMMARY OF FUNDING & PUBLICATIONS

Research Funding

- *Grant Record*: Secured **10+ competitive external grants** (e.g., NSF, DOD, Open Technology Fund, DOE, DARPA, Industry) as Lead PI (2016–2026), including
 - 7+ (**70%**) *since receiving tenure* (i.e., 2022)
- *Direct Funding (PI)*: Total funding raised as Lead/Institutional PI: **\$4.4M**, including
 - \$3.1M (**70%**) *since receiving tenure* (i.e., 2022)
- *Collaborative Impact (PI/Co-PI)*: Total aggregate funding involved: **\$13.5M**, including
 - \$11.1M (**82%**) *since receiving tenure* (i.e., 2022)

Research Publications

- *Top-tier journals*: 32 (e.g., ACM TOSEM, IEEE TSE/TIFS/TDSC, CSUR), including
 - 18 (56%) as first or corresponding author
 - 21 (**66%**) *since receiving tenure* (i.e., 2022)
- *Top-tier conferences*: 35 (also on CSRankings) conference papers (in USENIX Security, CCS, S&P, NDSS, ICSE, FSE, ISSTA, ASE), including
 - 23 (66%) as first or corresponding author
 - 30 (**86%**) *since receiving tenure* (i.e., 2022)

PROFESSIONAL EXPERIENCE (*Industry*)

Baidu Inc., Beijing, China

Technical Lead at the Portal Search Autotest Group

07/2009 - 08/2010

- Technical management, autotest framework design and foundation class library architecting, cross-team reconciliation, engineer recruiting and mentoring

Baidu Inc., Beijing, China

Software Development Engineer at the Portal Search Autotest Group

07/2008 - 06/2009

- Autotest system design and implementation, mock building for webpage crawler, general-purpose testing toolkit development

Hangzhou Lianzheng Info.&Tech. Co., Ltd., Hangzhou, Zhejiang, China

Software Contractor

02/2007 - 06/2007

- Independent development of hearing test and evaluation system running on an embedded medical instrument for self-service hearing diagnosis, especially for the disabled

Wellhope Info.&Tech. Co., Ltd., Shanghai, China

Software Development Engineer at the Information Security Group

03/2006 - 08/2006

- Cryptologic algorithm research and development, optimization and validation for the public key infrastructure (PKI) middleware libraries supporting e-government platform, online

certificate status protocol (OCSP) platform and instant messaging (IM) component development

SNDA Entertainment Co., Ltd., Shanghai, China

Software Contractor

10/2005 - 02/2006

- Independent development of online Legend game plug-ins for remote database access and control

Hubei Tri-Ring Info.&Tech. Co., Ltd., Wuhan, Hubei, China

Senior Software Development Engineer at Embedded Software Group

02/2005 - 10/2005

- Design and implementation of the system-level integrated software platform for commercial fiscal cash register (FCR), USB driver development for FCR

EDUCATION

The University of Notre Dame, Notre Dame, IN, USA

Ph.D. in Computer Science and Engineering, 2015

Thesis: Cost-effective Dependency Analysis for Reliable Software Evolution

Advisors: Dr. Raul Santelices and Dr. Douglas Thain

GPA: 4.0/4.0

The University of Southern Mississippi, Hattiesburg, MS, USA

M.S. in Computer Science, 2012

Thesis: A Scientific Visualization Language for Tensor Field Visualizations

GPA: 4.0/4.0

Zhejiang University, Hangzhou, China

M.E. in Computer Science and Technology, 2008

Thesis: Algorithms for Motion Tracking in Streaming Videos

GPA: 3.94/4.0

Wuhan University, Wuhan, China

B.E. in Computer Science, 2004

GPA: 90.5/100

AWARDS / HONORS

1. *Distinguished Reviewer Award*, ASE, 2025
2. *Distinguished Reviewer Award*, ISSTA, 2025
3. *Distinguished Reviewer Award*, FSE, 2025
4. *TOSEM Outstanding Paper Award*, ACM TOSEM, 2025 (This is the inaugural conferment of the TOP award given to one paper annually by the ACM TOSEM journal. It recognizes the most impactful journal first paper published five years ago (2019-20 papers were considered in 2025). This award is for my TOSEM paper on sustainable learning-based malware detection. The TOP award committee selected it with the quote: “*This award recognizes your paper’s pioneering introduction of the malware-detection sustainability concept and its clear impact in extending evaluation practices in mobile-security research.*” ACM SIGSOFT kindly conducted an interview with me on this work.).
5. *Distinguished Reviewer Award*, NDSS, 2025
6. *World’s Top 2% Scientist*, Stanford/Elsevier Database, 2023-2024 (Ranked Top 0.53% in my sub-field of Software Engineering/Security based on standardized citation indicators)
7. *Mid-Career Research Excellence Award*, EECS at WSU, 2024
8. *Distinguished Reviewer Award*, IEEE TSE, 2024
9. *Distinguished Reviewer Award*, NDSS, 2024
10. *Huie-Rogers Endowed Chair in Computer Science*, EECS at WSU, 2023
11. *World’s Top 2% Scientist*, Stanford/Elsevier Database, 2022-2023 (Ranked Top 0.58% in my sub-field of Software Engineering/Security based on standardized citation indicators)

12. *ACM SIGSOFT Distinguished Paper Award*, ISSTA, 2022
13. *VCEA Junior Faculty Research Award*, 2022 (one junior faculty is selected annually from the Voiland College of Engineering and Architecture (VCEA, which consists of 8 schools/departments) to receive this award)
14. *EECS Early Career Award*, 2022 (one junior faculty is selected annually from the School of Electrical Engineering and Computer Science (EECS) to receive this award)
15. *Ranked as one of the world's top 20 most active early-stage researchers in software engineering worldwide*, 2021 (according to the statistics based on papers published at top-quality software engineering venues)
16. *Distinguished Reviewer Award* (TOSEM), ACM Publications Board, 2020
17. *Best Graduate Research Award* (two years in a row), USM, 2011, 2012
18. *Outstanding Technology Award*, Baidu Inc., 2009
19. *Excellent Master Thesis Award*, Zhejiang University, 2008
20. *Outstanding Postgraduate Award*, Zhejiang University, 2007

FUNDING SUPPORT

(All PI/Co-PIs are
listed for each
project.)

Total amount as (sole/lead/institution) **PI: \$4.4 M**; Total funding involved: \$13.5M

External Funding

1. *SHF: Extreme Semantic Dynamic Analysis of Multilingual Code*
Agency: National Science Foundation (NSF CISE core programs/Future CoRe–CCF/SHF)
Period of performance: 01/01/27–12/30/29
Amount to my institution: **\$615,000**
Role: Sole PI
2. *CyberAI Innovation: Workforce Development for Integrated AI and Cybersecurity in Healthcare Systems*
Agency: National Science Foundation (NSF CyberAI program)
Period of performance: 01/01/27–12/30/29
Amount to my institution: **\$500,000**
Role: PI (with Co-PI Prof. Xu@UB)
Status: recommended on Research.gov (July 2026); official award anticipated summer/fall 2026
3. *CyberCorps Scholarship for Service: Developing the Next-generation Cyber Workforce*
Agency: National Science Foundation (NSF DGE)
Period of performance: 06/01/24–05/31/29
Amount to my institution: **\$3,575,646**; my share: \$35,130 (0.5 month salary per year, for 5 years—this is an education/training grant; the vast majority of the funds goes to student scholarship)
Role: Co-PI, focusing on software supply chain security (with PI Assefaw H Gebremedhin, and other Co-PIs Janardhan Rao Doppa, Feng-Hao Liu, and Monowar Hasan)
4. *URICS: Understanding Repressive Internet Communities via Mobile Software Ecosystem Mining*
Agency: Open Technology Fund (OTF Internet Freedom Fund)
Period of performance: 08/15/23–08/14/25
Amount to my institution: **\$250,000**
Role: Sole PI
5. *HS and UG Apprenticeships: Realistic Vulnerability Data Generation Using Large Language Models*
Agency: Army Educational Outreach Program (AEOP)
Period of performance: 6/1/2024–8/31/2024
Amount to my institution: **\$24,000**
Role: Sole PI

6. *REU Supplement: Practical Dynamic Program Reasoning Across Language Boundaries*
Agency: National Science Foundation (NSF CISE/CCF)
Period of performance: 06/01/23–05/31/24
Amount to my institution: **\$16,000**
Role: Sole PI
7. *HSAP/URAP: Real-World Software Vulnerability Mining and Validation*
Agency: Army Educational Outreach Program (AEOP)
Period of performance: 6/1/2023–8/31/2023
Amount to my institution: **\$12,000**
Role: Sole PI
8. *SHF: Small: Practical Dynamic Program Reasoning Across Language Boundaries*
Agency: National Science Foundation (NSF CISE core programs–CCF/SHF)
Period of performance: 06/01/22–05/30/27
Amount to my institution: **\$485,768**
Role: Sole PI
9. *HSAP/URAP: Assessing the Authenticity of Deep Learning Generated Software Vulnerabilities*
Agency: Army Educational Outreach Program (AEOP)
Period of performance: 6/1/2022–8/31/2022
Amount to my institution: **\$12,000**
Role: Sole PI
10. *Autonomic Defense of Distributed Information Security in Dynamic and Adversarial Environments*
Agency: Office of Naval Research (ONR Cyber Security and Complex Software Systems program)
Period of performance: 3/1/22–2/28/27
Amount to my institution: **\$561,710**
Role: Sole PI
11. *PMU: VICEROY Northwest Institute for CyberSecurity Education & Research (CySER) Sustainment Funding*
Agency: Department of Defense (DoD VICEROY program)
Period of performance: 09/01/2023 — 09/15/2025
Amount to my institution: **\$500,000**; my share: 2%
Role: Senior Personnel
(PI: Prof. Gebremedhin@WSU; Co-PIs: Prof. Van Wie@WSU, Prof. Schulz@WSU)
12. *Hierarchical Software Quality Assurance*
Agency: Department of Homeland Security (DHS) Science and Technology (S&T) Directorate
Period of performance: 10/1/2022–9/30/2026
Amount to my institution: **\$861,768** (total: \$4,500,000)
Role: Institution (WSU) PI
(with Prof. Izurieta@Montana State University as the lead PI, and other institution PI/Co-PIs: Prof. Arnaoudova@WSU, Prof. Opitz@University of Montana, Mr. McBride@Idaho State University, Prof. Mirakhorli@RIT)
13. *REU Supplement: Towards Self-Adaptive Dynamic Analysis of Distributed Software*
Agency: National Science Foundation (NSF CISE/CCF)
Period of performance: 08/01/21–07/31/22
Amount to my institution: **\$16,000**
Role: Sole PI

14. *Northwest Virtual Institute for CyberSecurity Education & Research (CySER)*
Agency: Department of Defense (DoD VICEROY program)
Period of performance: 06/01/2021 — 06/01/2025
Amount to my institution: **\$1,000,000** (total: \$1,500,000); my share: **\$97,367**
Role: Senior Personnel
(PI: Prof. Van Wie@WSU; Co-PIs: Prof. Gebremedhin@WSU, Prof. Schulz@WSU, Prof. Arnaoudova@WSU)
15. *Advancing the Knowledge about Software Vulnerability Analysis via Large-Scale Benchmarking*
Agency: Army Research Office (ARO Computer Science core program)
Period of performance: 2/1/2021–7/31/2025
Amount to my institution: **\$569,990**
Role: Lead PI
(with a subcontract of \$265,697 to Co-PI Prof. Chen@UT Dallas)
16. *Countermeasures against Information Flow Attack Surfaces via Multi-Phased Tainting*
Agency: Department of Energy via Pacific Northwest National Laboratory
Period of performance: 3/2/20–9/30/2021
Amount to my institution: **\$68,241**
Role: Sole PI
17. *Demonstrating Scalable Whole-System Code Analysis of HPC Software*
Agency: Department of Energy via Pacific Northwest National Laboratory
Period of performance: 08/07/19–4/30/20
Amount to my institution: **\$57,487**
Role: Sole PI
18. *SHF: EAGER: Towards Self-Adaptive Dynamic Analysis of Distributed Software*
Agency: National Science Foundation (NSF CISE core programs–CCF/SHF)
Period of performance: 08/01/19–07/31/22
Amount to my institution: **\$150,000**
Role: Sole PI
19. *Automatic Generation of Anti-Specifications from Exploits for Scalable Program Hardening*
Agency: DARPA (CASE seedling program)
Period of performance: 5/1/2018–9/27/2018
Amount to my institution: **\$20,000** (total \$400,000)
Role: Institution (WSU) PI
(with Prof. Yao@Virginia Tech as the lead PI and other institution PI/Co-PIs: Prof. Tan@PSU)

Internal Funding

1. *Improving the Design of the Software Testing Course*
Agency: Department of Education Higher Education Emergency Relief Fund (HEERF) via WSU Office of Provost (Course Redesign Grant)
Period of performance: 6/1/2021–5/30/2022
Amount: **\$4,000**
Role: Sole PI
2. *Incremental Taint Analysis for Efficient and Precise Vetting of Evolving Android Apps*
Agency: WSU Office of Research (New Faculty Seed Grant)
Period of performance: 5/1/2017–3/20/2020
Amount: **\$29,998**
Role: Sole PI
(9 out of 49 applications were funded)

PUBLICATIONS

Google scholar statistics: h-index 37, i10-index 103, total citations 4,350

(Students/(co-)advisees are underscored.)

(My name is tagged with * on papers where I am the corresponding author.)

(Venues of top-tier, representative papers are highlighted in *blue*.)

(For conference papers, acceptance rate (AR) of the respective conference is given when available.)

Peer-Reviewed Journal Articles [*since starting faculty career*]

1. Yu Nong, Guangbei Yi, Mohammed Aldeen, Long Cheng, Hongxin Hu, and Haipeng Cai*. 2026. Assessing and Improving Prompting Large Language Models for Software Vulnerability Analysis. *ACM Transactions on Software Engineering and Methodology* (**TOSEM**) 0(0), pp. 1–53. DOI: 10.1145/3821416. (impact factor: 6.9; journal-first paper).
2. Jiawei Guo, Yu Nong, Zhiqiang Lin, and Haipeng Cai*. 2026. FreeLens: Revealing Mobile App Feature Differences and Their Security and Privacy Implications across Geographical Regions. *IEEE Transactions on Dependable and Secure Computing* (**TDSC**) 0(0), pp. 1–18. DOI: 10.1109/TDSC.2026.3701655. (impact factor 7.5).
3. Deepika Shanmugasundaram, Pallavi Arivukkarsu, Huaming Chen, and Haipeng Cai*. 2025. Deep Learning Representations of Programs: A Systematic Literature Review. *ACM Computing Surveys* (**CSUR**) 58(5), pp. 1–37. DOI: 10.1145/3769008. (impact factor: 23.8).
4. Seyed Shayan Daneshvar, Yu Nong, Xu Yang, Shaowei Wang, and Haipeng Cai. 2025. VulScribeR: Exploring RAG-based Vulnerability Augmentation with LLMs. *ACM Transactions on Software Engineering and Methodology* (**TOSEM**) 0(0), pp. 1–24. DOI: 10.1145/3760775. (impact factor: 6.6; journal-first paper).
5. Haipeng Cai*. 2025. A Survey of Program Analysis for Distributed Software Systems. *ACM Computing Surveys* (**CSUR**) 0(0), pp. 1–41. DOI: 10.1145/3742900. (impact factor: 23.8).
6. Jiawei Guo and Haipeng Cai*. 2025. EvoTaint: Incremental Static Taint Analysis of Evolving Android Apps. *ACM Transactions on Software Engineering and Methodology* (**TOSEM**) 0(0), pp. 1–44. DOI: 10.1145/3743132. (impact factor: 6.6).
7. Iftexhar Ahmed, Aldeida Aleti, Haipeng Cai, Alexander Chatzigeorgiou, Pinjia He, Xing Hu, Mauro Pezzè, Denys Poshyvanyk, and Xin Xia. 2025. Artificial Intelligence for Software Engineering: The Journey so far and the Road ahead. *ACM Transactions on Software Engineering and Methodology* (**TOSEM**) 0(0), pp. 1–25. DOI: 10.1145/3719006. (impact factor: 6.6).
8. Jiawei Guo, Xiaoqin Fu, Li LI, Tao Zhang, Mattia Fazzini, and Haipeng Cai*. 2025. Characterizing Installation- and Run-Time Compatibility Issues in Android Benign Apps and Malware. *ACM Transactions on Software Engineering and Methodology* (**TOSEM**) 0(0), pp. 1–43. DOI: 10.1145/3725810. (impact factor: 6.6).
9. Yao Li, Sen Fang, Tao Zhang, and Haipeng Cai. 2025. Enhancing Android Malware Detection: The Influence of ChatGPT on Decision-centric Task. *ACM Transactions on Software Engineering and Methodology* (**TOSEM**) 0(0), pp. 1–30. DOI: 10.1145/3720541. (impact factor: 6.6).
10. Li LI, Xiang Gao, Hailong Sun, Chunming Hu, Xiaoyu Sun, Haoyu Wang, Haipeng Cai, Ting Su, Xiapu Luo, Tegawend Bissyande, Jacques Klein, John Grundy, Tao Xie, Haibo Chen, and Huamin Wang. 2025. Software engineering for openharmony: A research roadmap. *ACM Computing Surveys* (**CSUR**) 0(0), pp. 1–34. DOI: 10.1145/3720538. (impact factor: 23.8).
11. Xiaoqin Fu, Asif Zaman, and Haipeng Cai*. 2024. DistMeasure: A Framework for Run-Time Characterization and Quality Assessment of Distributed Software via Interprocess Communications. *ACM Transactions on Software Engineering and Methodology* (**TOSEM**) 0(0), pp. 1–52. DOI: 10.1145/3708476. (impact factor: 6.6).
12. Yao Li, Dawei Yuan, Tao Zhang, Haipeng Cai, David Lo, Cuiyun Gao, Xiapu Luo, and He Jiang. 2024. Meta-Learning for Multi-Family Android Malware Classification. *ACM Transactions on Software Engineering and Methodology* (**TOSEM**) 33(7), pp. 1–27. DOI: 10.1145/3664806. (impact factor: 6.6; journal-first paper).
13. Bo Li, HaoWei Quan, Jiawei Wang, Pei Liu, Haipeng Cai, Yuan Miao, Yun Yang, and Li LI. 2024. Neural Library Recommendation by Embedding Project-Library Knowledge Graph. *IEEE Transactions on Software Engineering* (**TSE**) 50(6), pp. 1620–1638. DOI: 10.1109/TSE.2024.3393504. (impact factor: 7.4; journal-first paper).

14. James Halvorsen, Clemente Izurieta, Haipeng Cai, and Assefaw Gebremedhin. 2024. Applying Generative Machine Learning to Intrusion Detection: A Systematic Mapping Study and Review. *ACM Computing Surveys (CSUR)* 56(10), pp. 1–33. DOI: 10.1145/3659575. (impact factor: 16.6).
15. Haoran Yang, Yu Nong, Shaowei Wang, and Haipeng Cai*. 2024. Multi-Language Software Development: Issues, Challenges, and Solutions. *IEEE Transactions on Software Engineering (TSE)* 50(03), pp. 512–533. DOI: 10.1109/TSE.2024.3358258. (impact factor: 7.4).
16. Wen Li, Austin Marino, Haoran Yang, Na Meng, Li LI, and Haipeng Cai*. 2023. How are Multilingual Systems Constructed: Characterizing Language Use and Selection in Open-Source Multilingual Software. *ACM Transactions on Software Engineering and Methodology (TOSEM)* 33(3). DOI: 10.1145/3631967. (impact factor: 6.6; journal-first paper, presented at ICSE 2024).
17. Yao Li, Tao Zhang, Xiapu Luo, Haipeng Cai, Sen Fang, and Dawei Yuan. 2023. Do Pre-trained Language Models Indeed Understand Software Engineering Tasks? *IEEE Transactions on Software Engineering (TSE)* 49(10), pp. 4639–4655. DOI: 10.1109/TSE.2023.3308952. (impact factor: 7.4; journal-first paper).
18. Pei Liu, Yanjie Zhao, Mattia Fazzini, Haipeng Cai, John Grundy, and Li LI. 2023. Automatically Detecting Incompatible Android APIs. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, pp. 1–33. DOI: 10.1145/3624737. (impact factor: 6.6).
19. Xiaoyu Sun, Xiao Chen, Li LI, Haipeng Cai, John Grundy, Jordan Samhi, Tegawende Bissyande, and Jacques Klein. 2022. Demystifying Hidden Sensitive Operations in Android apps. *ACM Transactions on Software Engineering and Methodology (TOSEM)* 32(2). DOI: 10.1145/3574158. (impact factor: 4.4; journal-first paper, presented at ESEC/FSE 2023).
20. Yu Nong, Rainy Sharma, Wahab Hamou-Lhadj, Xiapu Luo, and Haipeng Cai*. 2022. Open Science in Software Engineering: A Study on Deep Learning-Based Vulnerability Detection. *IEEE Transactions on Software Engineering (TSE)* 49(4), pp. 1983–2005. DOI: 10.1109/TSE.2022.3207149. (impact factor: 9.322; journal-first paper, presented at ICSE 2023).
21. Haipeng Cai* and Xiaoqin Fu. 2021. D²ABS: A Framework for Dynamic Dependence Analysis of Distributed Programs. *IEEE Transactions on Software Engineering (TSE)* 48(12), pp. 4733–4761. DOI: 10.1109/TSE.2021.3124795. (impact factor: 6.226).
22. Yu Nong, Haipeng Cai*, Pengfei Ye, Li LI, and Feng Chen. 2021. Evaluating and Comparing Memory Error Vulnerability Detectors. *Information and Software Technology (IST)* 137, pp. 106614. DOI: 10.1016/j.infsof.2021.106614. (impact factor 2.726).
23. Long Cheng, Salman Ahmed, Hans Liljestrand, Thomas Nyman, Haipeng Cai, Trent Jaeger, Nadarajah Asokan, and Danfeng Yao. 2021. Exploitation Techniques for Data-Oriented Attacks with Existing and Potential Defense Approaches. *ACM Transactions on Privacy and Security (TOPS)* 24(4), pp. 1–36. DOI: 10.1145/3462699. (impact factor: 1.974).
24. Yanjie Zhao, Li LI, Haoyu Wang, Haipeng Cai, Tegawende Bissyande, Jacques Klein, and John Grundy. 2021. On the Impact of Sample Duplication in Machine Learning based Android Malware Detection. *ACM Transactions on Software Engineering and Methodology (TOSEM)* 30(3), pp. 1–38. DOI: 10.1145/3446905. (impact factor 3.7; journal-first paper, presented at ESEC/FSE 2022).
25. Xiaoqin Fu, Haipeng Cai*, Wen Li, and Li LI. 2020. Seeds: Scalable and Cost-Effective Dynamic Dependence Analysis of Distributed Systems via Reinforcement Learning. *ACM Transactions on Software Engineering and Methodology (TOSEM)* 30(1), pp. 1–45. DOI: 10.1145/3379345. (impact factor 2.7; journal-first paper, presented at ESEC/FSE 2021).
26. Haipeng Cai*, Xiaoqin Fu, and Abdelwahab Hamou-Lhadj. 2020. A Study of Run-time Behavioral Evolution of Benign versus Malicious Apps in Android. *Information and Software Technology (IST)* 122, pp. 106291. DOI: 10.1016/j.infsof.2020.106291. (impact factor 2.726; journal-first paper, presented at ICSME 2020).
27. Haipeng Cai* and Barbara Ryder. 2020. A Longitudinal Study of Application Structure and Behaviors in Android. *IEEE Transactions on Software Engineering (TSE)* 47(12), pp. 2934–2955. DOI: 10.1109/TSE.2020.2975176. (impact factor 6.11).
28. Haipeng Cai*. 2020. Assessing and Improving Malware Detection Sustainability through App Evolution Studies. *ACM Transactions on Software Engineering and Methodology (TOSEM)* 29(2), pp. 8:1–8:28. DOI: 10.1145/3371924. (impact factor 2.7; journal-first paper, presented at ESEC/FSE 2020; **TOSEM Outstanding Paper Award**).

29. Sazzadur Rahaman, Haipeng Cai, Omar Chowdhury, and Danfeng Yao. 2019. From Theory to Code: Identifying Logical Flaws in Cryptographic Implementations in C/C++. *IEEE Transactions on Dependable and Secure Computing (TDSC)* 19(6), pp. 3790–3803. DOI: 10.1109/TDSC.2021.3108031. (impact factor 6.4).
30. Karim Elish, Haipeng Cai, Daniel Barton, Danfeng Yao, and Barbara Ryder. 2018. Identifying Mobile Inter-App Communication Risks. *IEEE Transactions on Mobile Computing (TMC)* 19(1), pp. 90–102. DOI: 10.1109/TMC.2018.2889495. (impact factor 4.8).
31. Haipeng Cai*, Na Meng, Barbara Ryder, and Danfeng Yao. 2019. DroidCat: Effective Android Malware Detection and Categorization via App-Level Profiling. *IEEE Transactions on Information Forensics and Security (TIFS)* 14(6), pp. 1455–1470. DOI: 10.1109/TIFS.2018.2879302. (impact factor 6.2; journal-first paper).
32. Nasir Ali, Haipeng Cai, Abdelwahab Hamou-Lhadj, and Jameleddine Hassine. 2019. Exploiting Parts-of-Speech for Effective Automated Requirements Traceability. *Information and Software Technology (IST)* 106, pp. 126–141. DOI: 10.1016/j.infsof.2018.09.009. (impact factor 2.627; journal-first paper).
33. Haipeng Cai*. 2018. Hybrid Program Dependence Approximation for Effective Dynamic Impact Prediction. *IEEE Transactions on Software Engineering (TSE)* 44(4), pp. 334–364. DOI: 10.1109/TSE.2017.2692783. (impact factor 6.11).
34. Li LI, Daoyuan Li, Tegawend Bissyande, Jacques Klein, Haipeng Cai, David Lo, and Yves Le Traon. 2017. On Locating Malicious Code in Piggybacked Android Apps. *Journal of Computer Science and Technology (JCST)* 32(6), pp. 1108–1124. DOI: 10.1007/s11390-017-1786-z. (impact factor 1.506).
35. Haipeng Cai* and Raul Santelices. 2016. Method-Level Program Dependence Abstraction and Its Application to Impact Analysis. *Journal of Systems and Software (JSS)* 122, pp. 311–326. DOI: 10.1016/j.jss.2016.09.048. (impact factor: 2.450).

Peer-Reviewed Journal Articles [before faculty career]

36. Haipeng Cai*, Raul Santelices, and Douglas Thain. 2016. DiaPro: Unifying Dynamic Impact Analyses for Improved and Variable Cost-Effectiveness. *ACM Transactions on Software Engineering and Methodology (TOSEM)* 25(2), pp. 18:1–18:50. DOI: 10.1145/2894751. (impact factor 2.516).
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123. Siyuan Jiang, Raul Santelices, Haipeng Cai, and Mark Grechanik. 2014. How Accurate Is Dynamic Program Slicing? An Empirical Approach to Compute Accuracy Bounds. In: *International Conference on Software Security and Reliability-Companion (SERE-C)*, pages 3–4. DOI: 10.1109/SERE-C.2014.14.
124. Jian Chen, Haipeng Cai, AlexanderP. Auchus, and DavidH. Laidlaw. 2014. “Gryphon: A Little Domain-Specific Programming Language for Diffusion MRI Visualizations”. In: *Handbook of Human Centric Visualization*. Ed. by Huang, Weidong. Springer New York, pp. 41–61. ISBN: 978-1-4614-7484-5. DOI: 10.1007/978-1-4614-7485-2_2.
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PHD STUDENTS

Current Students

1. **Zhengsong Zhang**, University at Buffalo, NY Fall 2025 -
research on software vulnerability analysis and mitigation
(**UB Presidential Fellowship**)
2. **Zongze Li**, University at Buffalo, NY Fall 2025 -
research on software and supply chain security of machine learning systems
(**UB CSE Chair Fellowship**)
3. **Rupam Patir**, University at Buffalo, NY Fall 2024 -
research on secure code generation with AI models (co-advisor with Prof. Hongxin Hu as main advisor)
4. **Jiawei Guo**, Washington State University, WA Fall 2023 -
(transferred to University at Buffalo with me in Fall 2024)
research on security of mobile software systems, frameworks, and applications
(**ACM SIGSOFT Travel Grant for ISSTA, 2025**)

Past Students

1. **Yu Nong**, Washington State University and University at Buffalo, WA Fall 2020 - Spring 2026
(transferred to University at Buffalo with me in Fall 2024)
research on data quality problems in data-driven software vulnerability analysis
PhD Dissertation title: *Understanding and Tackling the Data Needs for Software Vulnerability Analysis* (passed defense in April 2026)
(**Best RA Award of EECS at WSU, 2024**)
(**S&P'25 Travel Grant, 2025**)
(**USENIX Security'25 Travel Grant, 2025**)
(graduated; first job: **tenure-track assistant professor in CS** at Oakland University, USA)
2. **Haoran Yang**, Washington State University, WA Fall 2021 - Fall 2025
research on multi-language software defect diagnosis based on deep learning
PhD Dissertation title: *Understanding and Detecting Defects in Multi-Lingual Software* (passed defense in December 2025)
(**ACM SIGSOFT Travel Grant for FSE, 2025**)
(graduated; first job: **tenure-track assistant professor in CS** at Sichuan University, China)
3. **Wen Li**, Washington State University, WA Fall 2019 - May 2024
research on multi-language software analysis and security
PhD Dissertation title: *Run-Time Analysis and Security of Multi-Language Software* (passed defense in December 2023)
(**Best RA Award of EECS at WSU, 2023**)
(graduated; first job: **tenure-track assistant professor in CS** at Utah State University, USA)

4. **Xiaoqin Fu**, Washington State University, WA Fall 2017 - Spring 2022
research on distributed software system analysis and security
PhD Dissertation title: *Scalable and Cost-Effective Data Flow Analysis for Distributed Software: Algorithms and Applications* (passed defense in May 2022)
(ACM SIGSOFT travel grant, 2019; **Mahmoud M.Dillsi Family Graduate Fellowship, 2020**)
(graduated; first job: research engineer at JDC Technology Inc.)
5. **Asif Zaman**, Washington State University, WA Fall 2022 - Spring 2024
research on distributed system analysis/security with machine/deep learning
(could not complete PhD due to personal reasons, graduated with an MS/thesis degree)
6. **Sanjeev Arora**, Washington State University, WA Spring 2022 - Summer 2022
part-time student, official employee at Microsoft; research on distributed system security
(currently on leave back full-time at Microsoft due to family financial needs)
7. **Shiv Raj Pant**, Washington State University, WA Spring 2019 - Summer 2019
research on vision-based code semantics modeling for program understanding and search
(could not complete PhD due to severe health issues, returned to Far Western University, Nepal as an assistant professor there)

MASTERS STUDENTS

Current Students

1. **Suvadra Barua**, University at Buffalo, NY Fall 2025 -
study on secure AI-driven code generation
2. **Johnathan Tang**, University at Buffalo, NY Spring 2026 - Spring 2026
study on time-sensitive vulnerability analysis
3. **Kefan Chen**, University at Buffalo, NY Spring 2026 -
study on time-sensitive vulnerability analysis
4. **Bhavin Gaikwad**, University at Buffalo, NY Summer 2026 -
study on time-sensitive vulnerability analysis
5. **Jash Vekariya**, University at Buffalo, NY Summer 2026 -
study on time-sensitive vulnerability analysis
6. **Krishna Meda**, Texas A&M University, TX Summer 2026 -
study on LLM agents for software/systems security
7. **Jingjing Nie**, Washington State University, WA Spring 2026 -
study on LLM agents for software/systems security

Past Students

1. **Abhijeet Pathak**, University at Buffalo, NY Fall 2025 - Spring 2026
study on secure AI-driven code generation
2. **Dinesh Gudimetla**, University at Buffalo, NY Fall 2025 - Spring 2026
study on secure AI-driven code generation
3. **Maihan Wen**, University at Buffalo, NY Fall 2024 - Fall 2025
research on distributed system security
4. **Andrew Dempsey**, University at Buffalo, NY Fall 2025 - Spring 2026
research on distributed system security
5. **Deepika Shanmugasundaram** (remote advisee, enrollment postponed), Washington State University, WA Spring 2024 - Summer 2024
research on surveying program representation learning and LLMs for software security
6. **Dheeraj Vurukuti** (co-advised with Dr. Jana Doppa), Washington State University, WA Fall 2023 - Fall 2024
research on sustainable and adaptive malware analysis

7. **Shlok Tomar** (co-advised with Dr. Jana Doppa), Washington State University, WA Fall 2023 - Fall 2024
research on code quality through large language models
 8. **Asif Zaman**, Washington State University, WA Fall 2022 - Spring 2024
research on distributed system analysis/security with machine/deep learning
 9. **Pallavi Arivukkarasu**, Washington State University, WA Spring 2021 - Spring 2023
study on program representation learning and its applications
(graduated; now SDE@Truveta)
 10. **Jinyang Ruan**, Washington State University, WA Fall 2021 - Spring 2022
research on greybox fuzzing of multi-language software
(graduated; now Graduate Analyst Developer at FNZ Group)
 11. **Rainy Sharma**, Washington State University, WA Fall 2020 - Spring 2021
research on open science in software engineering
(graduated; now a QA engineer at Matilda Cloud Solutions)
 12. **Weile Lian**, Washington State University, WA Spring 2019–Fall 2020
research on study of cross-language software development issues
(graduated; now SDE@Ruiyun Network Technology)
 13. **John Jenkins**, Washington State University, WA Spring 2017 - Fall 2020
research on program understanding and security for mobile apps
Thesis title: *Incremental Taint Analysis of Android Applications*
(graduated, now Software Engineer at Schweitzer Engineering Laboratories)
 14. **Yu Nong**, Washington State University, WA Spring 2019 - Spring 2020
research on comparing software vulnerability detectors
Thesis title: *An Empirical Study of Memory Vulnerability Detection Techniques*
(graduated, now Ph.D. student at WSU)
 15. **Ziyi Zhang**, Washington State University, WA Fall 2017 - Spring 2019
research on compatibility issues in Android apps
(graduated, now Ph.D. student at WSU)
- (And 50+ non-thesis masters students, all graduated.)

UNDERGRADUATE, Current Students

HIGH SCHOOL &
INTERN
STUDENTS
INVOLVED IN
RESEARCH

1. **Leihan Wang**, University at Buffalo, NY Spring 2025 - Spring 2026
prompt optimization for LLM code generation

Past Students

1. **Yan Zhang**, REU student, Ohio State University, OH Fall 2024 - Spring 2025
research on large language models for cross-language static code analysis
2. **Colin Xiao**, REU student, University of Washington at Seattle, WA Spring 2024 - Fall 2025
research on large language models for automated developer discussion analysis
3. **Jason Zhang**, REU student, University of Maryland at College Park, MD Spring 2024 - Fall 2025
research on large language models for automated developer discussion analysis
4. **Haridhar Pulivarthi**, REU student, Washington State University, WA Summer 2023 - Spring 2024
research on LLM-based data augmentation for software vulnerability analysis
5. **Richard Fang**, REU student, University of Illinois Urbana-Champaign, IL Fall 2023 - Spring 2024
research on LLM-based data augmentation for software vulnerability analysis
6. **Pradyumna Kotla**, REU student, University of Maryland, MD Summer 2023
research on security of LLM-generated code

7. **Guangbei Yi**, REU student, Washington State University, WA Summer 2022, Fall 2022, Spring 2023, Fall 2024—Fall 2025
research on cross-language fuzzing and data generation for software vulnerability analysis
third author as an undergrad student on a top SE conference paper (ICSE'24 full research paper[60]) and a top security conference paper (USENIX Security'23 full research paper[65]), and first author on a top SE conference paper (ICSE'26 full research paper[46])
(Undergraduate Research Award at WSU, 2024)
8. **Bella Fang**, REU student, Washington State University, WA Summer 2022
research on data generation for software vulnerability analysis
9. **Richard Fang**, high-school student, Washington State University, WA Summers 2022, 2023
research on pattern-based data generation and LLM-based data augmentation for software vulnerability analysis
second author as a high-school student on a top SE conference paper (ICSE'24 full research paper[60])
10. **Boxiang Lin**, REU student, Washington State University, WA Fall 2021 - Spring 2022
research on Android application compatibility issues
second author as an undergrad on a top SE conference paper (ICSE'22 tool demo[76])
11. **Megan Jung**, undergraduate researcher, Cornell University, NY Fall 2020
research on multi-language software development issues
12. **Alissa Cielecki**, REU student, Arcadia University, PA Summer 2020
research on multi-language software vulnerabilities
(graduated, now Tech Specialist at the Wharton School)
13. **Austin Marino**, REU student, Washington State University, WA Summer 2019
research on multi-language use in open-source software projects
second author as an undergrad on a top SE journal paper (TOSEM'23[16])
(graduated, now Software Engineer at General Motors)
14. **Brandon Campbell**, undergraduate researcher, Washington State University, WA Fall 2018
research on incremental taint analysis for mobile apps
(graduated, now Software Engineer at Capital One)
15. **Linh Nguyen**, undergraduate researcher, Washington State University, WA Fall 2017
research on prioritized precise taint analysis for mobile apps
16. **Sophya Wu**, high school student researcher, Pullman High School, WA Summer 2019
research on multi-language use in open-source software projects
(graduated, now CS undergraduate student at Columbia University)
17. **Malinda Dilhara**, remote undergraduate researcher, Sri Lanka Spring 2017 - Spring 2018
research on adapting mobile software for runtime permission models
first author as an undergrad on a research paper (MOBILESoft'18[95]), who also presented the paper at the conference
(graduated, now CS Ph.D. student at CU Boulder)

TEACHING EXPERIENCE

At UB

1. Instructor, *CSE713 Software Analysis and Applications*, University at Buffalo Spring 2026
2. Instructor, *CSE418/518 Software Security*, University at Buffalo Spring 2025, Fall 2025
3. Instructor, *CSE410/518 Software Security*, University at Buffalo Fall 2024

At WSU

1. Instructor, *CptS428/528 Advanced Cyber Security*, Washington State University Fall 2022, Fall 2023
2. Course Designer, *CptS428/528 Advanced Cyber Security*, Washington State University Fall 2021-Summer 2022

3. Instructor, *CptS583 Software Quality*, Washington State University Fall 2018, Spring 2020-2022
4. Instructor, *CptS422 Software Engineering Principles II*, Washington State University Fall 2018-2022
5. Instructor, *CptS322 Software Engineering Principles I*, Washington State University Fall 2016, Spring 2017-2024
6. Course Designer, *CptS583 Software Quality*, Washington State University Fall 2017

Table 1: Teaching assignments and students' rating scores

Course #	Title	Semester	Enrollment	Evaluation score (out of 5)		
				course	instructor	overall
CptS322	Software Engineering Principles I (at WSU)	Fall 2016	61	2.3	2.4	2.3
		Spring 2017	72	3.8	4.0	3.9
		Spring 2018	62	4.1	4.3	4.2
		Spring 2019	95	4.3	4.3	4.3
		Spring 2020	101	4.3	4.5	4.4
		Spring 2021	125	4.4	4.6	4.5
		Spring 2022	123	4.1	4.3	4.2
		Spring 2023	121	3.9	4.1	4.0
		Spring 2024	123	3.4	3.4	3.4
CptS422	Software Engineering Principles II (at WSU)	Fall 2018	76	3.9	4.1	4.0
		Fall 2019	48	3.9	4.0	4.0
		Fall 2020	46	4.3	4.5	4.4
		Fall 2021	33	4.5	4.8	4.6
		Fall 2022	33	3.7	3.8	3.8
CptS583	Software Quality (at WSU)	Fall 2018	6	4.3	4.3	4.3
		Spring 2020	28	4.9	5.0	4.9
		Spring 2021	32	4.7	5.0	4.8
		Spring 2022	12	4.6	4.4	4.5
CptS428/528	Advanced Cyber Security (Software Security and Reverse Engineering) (at WSU)	Fall 2022	12	4.3	4.9	4.6
		Fall 2023	39	4.4	4.4	4.4
CSE410/518	Software Security (at UB)	Fall 2024	39	3.75	4.0	3.9
CSE418/518	Software Security (at UB)	Spring 2025	31	3.8	4.3	4.1
		Fall 2025	38	3.6	4.1	3.8
CSE713	Software Analysis and Applications (at UB)	Spring 2026	10	5.0	5.0	5.0
Average		All semesters	1,366	4.09	4.27	4.18
		Since tenure (2022-)	581	4.05	4.25	4.15
		At UB (2024-)	118	4.04	4.35	4.20

Prior to WSU

1. Guest Lecturer, *CS5314 Concepts of Programming Languages*, Virginia Tech Spring 2016
2. Teaching Assistant, *CSE40613/60613 Web Programming*, Notre Dame Spring 2013
3. Guest Lecturer, *CSC333 Problem Solving in C*, USM Fall 2011
4. Teaching Assistant, *CSC333 Problem Solving in C*, USM Spring 2011
5. Guest Lecturer, *CS414 Software Design and Development*, USM Fall 2010
6. Instructor, *C/C++ Programming*, Gutian Computer Training School, Wuhan Summer 2003
7. Teacher, *Junior High School Mathematics*, Zhuwa Middle School, Hubei 01/2000 - 07/2002

MENTORING EXPERIENCE

At UB

- **20 CSE undergraduate students per year**, University at Buffalo 2024 - present
- mentoring on course selection, career options, graduate study pursuit, internship opportunities, and research experience opportunities as undergraduates

At WSU

- **26–38 CS undergraduate students per year**, Washington State University 2016 - 2024 mentoring on course selection, career options, graduate study pursuit, internship opportunities, and research experience opportunities as undergraduates

Prior to WSU

1. **Fang Liu**, graduate student, Virginia Tech, VA 2015
research on prioritized risk analysis of inter-component communication risks in Android
2. **Frank Cipollone**, REU student, University of Notre Dame, IN 2014
research on supporting program understanding via code structure visualization
3. **Tianyu Xu**, REU student, Fudan University, China 2013
research on evaluating dynamic change impact analysis techniques
4. **Jack Magiera**, REU student, University of Notre Dame, IN 2013
research on software repository mining to support descriptive change impact analysis
5. **12 Software Engineers**, Baidu, China 2009-2010
development of a series of search-engine automated testing tools

PROFESSIONAL SERVICES

Conference Organizing Committee Membership

1. Journal First Co-Chair, IEEE/ACM International Conference on Software Engineering (ICSE), 2025
2. Program Vice Co-Chair, IEEE International Conference on Internet of Things (iThings), 2024
3. Web Chair, Annual Workshop on Cyber Security in High Performance Computing (S-HPC), 2024
4. Regional Co-Chair, Annual Workshop on Cyber Security in High Performance Computing (S-HPC), 2023
5. General Co-Chair, Annual Workshop on Cyber Security in High Performance Computing (S-HPC), 2022
6. Program Co-chair, International Conference on Mobile Software Engineering and Systems (MOBILESoft) – Tools and Datasets track, 2023
7. Program Co-chair, International Conference on Mobile Software Engineering and Systems (MOBILESoft) – NIER track, 2022
8. Program Co-chair, International Conference on Mobile Software Engineering and Systems (MOBILESoft) - SRC track, 2020
9. Program Co-chair, International Conference on Mobile Software Engineering and Systems (MOBILESoft) - SRC track, 2019

Conference Technical Program Committee Membership

1. USENIX Security Symposium Award Committee, 2026
2. USENIX Security Symposium, 2026, 2027
3. IEEE Symposium on Security and Privacy (S&P), 2026, 2027
4. ACM Conference on Computer and Communications Security (CCS), 2023, 2024, 2025, 2026
5. ISOC Network and Distributed System Security Symposium (NDSS), 2023, 2024, 2025, 2026, 2027
6. IEEE/ACM International Conference on Software Engineering (ICSE), 2025, 2026, 2027
7. ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), 2023, 2024, 2025, 2026, 2027
8. ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2024, 2025, 2026
9. IEEE/ACM International Conference on Automated Software Engineering (ASE), 2019, 2024, 2025, 2026
10. ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) – Ideas, Visions and Reflections, 2022, 2023

11. AAAI Conference on Artificial Intelligence (AAAI) Senior PC, 2023, 2024, 2025, 2026, 2027
12. AAAI Conference on Artificial Intelligence (AAAI) Student Program, 2024
13. IEEE/ACM International Conference on Software Engineering (ICSE) – SRC track, 2024
14. IEEE/ACM International Conference on Software Engineering (ICSE) – Poster track, 2023
15. International Conference on Evaluation and Assessment in Software Engineering (EASE) – Vision and Emerging Results, 2023, 2024
16. International Conference on Internetware (InternetWare), 2025
17. International Workshop on Future of Software Engineering (FSEws), 2025
18. Asia-Pacific Software Engineering Conference (APSEC) – Early Research Achievements track, 2024
19. International Conference on Mobile Software Engineering and Systems (MOBILESoft) – Vision track, 2023
20. International Conference on Mobile Software Engineering and Systems (MOBILESoft) – Tool Demos and Mobile Apps track, 2022
21. European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2020
22. ACM Student Research Competition (ACM SRC), Judge at ICSE, 2024
23. ACM Student Research Competition (ACM SRC), Grand Finals judge committee, 2019, 2020, 2021
24. EAI International Conference on Security and Privacy in New Computing Environments (SPNCE), 2020, 2021, 2022
25. IEEE Secure Development Conference (SecDev), 2018, 2025
26. IEEE International Conference on Software Analysis, Evolution, and Reengineering (SANER) – ERA track, 2020
27. International Conference on Software Maintenance and Evolution (ICSME) – tool track, 2017, 2018
28. IEEE/ACM International Conference on Program Comprehension (ICPC) – ERA track, 2020
29. IEEE/ACM International Conference on Program Comprehension (ICPC) – Negative Results track, 2019
30. International Workshop on Software Security from Design to Deployment (SEAD), collocated with ESEC/FSE, 2019, 2020
31. International Workshop on Advances in Mobile App Analysis (A-Mobile), collocated with ASE, 2018, 2019, 2020, 2021, 2022
32. International Symposium on Advanced Security on Software and Systems (ASSS 2022), collocated with EuroS&P, 2022
33. International Workshop on Program Debugging (IWPD), 2015, 2016, 2017
34. International Conference on Software Engineering Advances (ICSEA), 2015, 2016, 2017, 2018, 2019
35. International Conference on Virtual Reality and Visualization (ICVRV), 2016
36. International Conference on Software Engineering and Applications (SEA), 2017
37. Third International Conference on Software Engineering (SOENG), 2017
38. Third International Conference on Software Security (ICSS), 2017
39. International Conference on Fundamentals and Advances in Software Systems Integration (FASSI), 2015, 2016

Conference External Reviewer

1. IEEE International Conference on Software Analysis, Evolution, and Reengineering (SANER), 2020
2. Annual Computer Security Applications Conference (ACSAC), 2019

3. ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec), 2017
4. ACM Conference on Computer and Communications Security (CCS), 2016
5. Annual Computer Security Applications Conference (ACSAC), 2016
6. IEEE Conference on Communications and Network Security (CNS), 2016
7. ACM Symposium on Information, Computer and Communications Security (ASIACCS), 2016
8. ACM Conference on Data and Application Security and Privacy (CODASPY), 2015
9. International Conference on Software Maintenance and Evolution (ICSME) - Tool Track, 2015
10. International Computers, Software & Applications Conference (COMPSAC), 2014
11. International Workshop on Program Debugging (IWPD), 2012

Journal Board Membership

1. Associate Editor, ACM Transactions on Software Engineering and Methodology (TOSEM), 2023-present
2. Associate Editor, Frontiers in Computer Science (Computer Security section), 2023-present
3. Associate Editor, Expert Systems with Applications (ESWA), 2022-present
4. Editorial Board Member, MDPI/Software (Software), 2024-present
5. Board of Distinguished Reviewers, ACM Transactions on Software Engineering and Methodology (TOSEM), 2019-2023
6. Review Board, Automated Software Engineering (AUSE), 2021-present
7. Guest Editor, Frontiers in Computer Science (*Computer Security* section), special edition on mobile security, 2021-2023
8. Guest editor, MDPI/Symmetry, Computer Science section, special edition on sustainable security for mobile apps, 2022-present

Journal Referee Publons statistics

1. ACM Transactions on Software Engineering and Methodology (TOSEM), 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026
2. IEEE Transactions on Software Engineering (TSE), 2018, 2019, 2021, 2022, 2023, 2024, 2025, 2026
3. IEEE Transactions on Information Forensics and Security (TIFS), 2017, 2020, 2021, 2022, 2023, 2024
4. IEEE Transactions on Dependable and Secure Computing (TDSC), 2016, 2017, 2020, 2021, 2022, 2024, 2025, 2026
5. ACM Transactions on Privacy and Security (TOPS), 2024, 2025, 2026
6. IEEE Transactions on Reliability (TRE), 2016, 2018, 2019, 2023
7. IEEE Transactions on Parallel and Distributed Systems (TPDS), 2017, 2018, 2019
8. IEEE Transactions on Network and Service Management (TNSM), 2021, 2023
9. IEEE Transactions on Knowledge and Data Engineering (TKDE), 2021
10. IEEE Transactions on Services Computing (TSC), 2022
11. IEEE Transactions on Computers (TC), 2023
12. ACM Transactions on Sensor Networks (TOSN), 2021
13. IEEE Transactions on Network Science and Engineering (TNSE), 2022, 2023
14. IEEE Transactions on Sustainable Computing (TSUSC), 2022
15. IEEE Transactions on Industrial Cyber-Physical Systems (TICPS), 2023
16. IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI), 2023
17. ACM Computing Surveys (CSUR), 2021, 2022, 2023
18. ACM Digital Threats: Research and Practice (DTRAP), 2019, 2020, 2021, 2022
19. Information and Software Technology (IST), 2018

20. Software: Practice and Experience (SPE), 2022, 2023
21. Software Quality Journal (SQJ), 2023
22. Journal of Software: Evolution and Process (JSME), 2022, 2023
23. Automated Software Engineering (AUSE), 2020, 2021, 2022, 2023, 2024
24. Empirical Software Engineering (EMSE), 2016, 2017, 2023, 2024
25. Computers & Security (COSE), 2019, 2021, 2022, 2023
26. IEEE Security & Privacy (SP), 2024
27. Transactions on Emerging Telecommunications Technologies (ETT), 2022, 2023
28. Journal of Computer Virology and Hacking Techniques (JICV), 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023
29. Journal of King Saud University - Computer and Information Sciences (JKSUCIS), 2018, 2020, 2021, 2022, 2023
30. Journal of Information Science and Engineering (JISE), 2018, 2020
31. Journal of Information Security and Applications (JISAS), 2020, 2021, 2023
32. Journal of Systems and Software (JSS), 2016, 2017, 2019, 2020, 2021
33. Journal of Parallel and Distributed Computing (JPDC), 2021
34. Journal of Intelligent & Fuzzy Systems (IFS), 2022
35. Journal of Computer Security (JCS), 2022
36. Journal of Computer Science and Technology (JCST), 2023
37. Journal of Engineering Research (JER), 2024
38. Frontiers of Information Technology & Electronic Engineering (FITEE), 2017
39. Human-centric Computing and Information Sciences (HCIS), 2018, 2019
40. Expert Systems with Applications (ESWA), 2018, 2019, 2020, 2021, 2022, 2023
41. Knowledge and Information Systems (KAIS), 2017
42. IEEE Access, 2018, 2019, 2020, 2021, 2023
43. Security and Communication Networks (SCN), 2020, 2021, 2022, 2023
44. Mobile Information Systems (MIS), 2020, 2021
45. IET Image Processing (IPR), 2020
46. IET Networks (NET), 2023
47. IET Software (SEN), 2023
48. IET Communications (COM), 2023, 2024
49. Signal Processing Letters (SPL), 2020
50. Concurrency and Computation: Practice and Experience (CPE), 2021, 2022, 2023
51. Wireless Communications and Mobile Computing (WCMC), 2019, 2020, 2021, 2022, 2023
52. Soft Computing (SOCO), 2021
53. Jordanian Journal of Computers and Information Technology (JJCIT), 2019, 2020, 2023
54. Chinese Journal of Electronics (CJE), 2022
55. International Journal of Automation and Computing (IJAC), 2021
56. The Journal of Supercomputing (SUPE), 2021
57. Journal of Combinatorial Optimization (JOCO), 2023
58. IEEE Systems Journal (ISJ), 2021, 2022, 2023
59. Wiley/Security and Privacy (SPY), 2021, 2022, 2023, 2024
60. Wiley/Computational Intelligence (COIN), 2021, 2022, 2023
61. Computational Intelligence and Neuroscience (CIN), 2022, 2023
62. Applied Computational Intelligence and Soft Computing (ACISC), 2022, 2023
63. IEEE Canadian Journal of Electrical and Computer Engineering (CJECE), 2023
64. Cybernetics and Systems (UCBS), 2023

65. IEEE Network Magazine, 2021
66. Big Data (BIG), 2022, 2023
67. IT Professional (ITPro), 2023
68. SoftwareX, 2023
69. Big Data Mining and Analytics, 2023
70. Blockchain: Research and Applications (BCRA), 2023
71. Formal Aspects of Computing (FAC), 2023, 2024
72. Expert Systems (EXSY), 2023
73. MDPI/Symmetry, 2019, 2020, 2022
74. MDPI/Sensors, 2019, 2020
75. MDPI/Entropy, 2019, 2020, 2023
76. MDPI/Applied Sciences, 2021, 2023
77. MDPI/Data, 2021
78. MDPI/Electronics, 2021
79. MDPI/Information, 2021
80. MDPI/Informatics, 2023
81. Medicine (MD), 2023

Journal Reviewer

1. Computers & Security (COSE), 2015
2. The Computer Journal (COMPJ), 2014
3. Software Quality Journal (SQJ), 2014
4. IEEE Transactions on Reliability (TRE), 2014
5. Science of Computer Programming (SCP), 2014
6. Journal of Object Technology (JoT), 2013
7. Software Testing, Verification and Reliability (STVR), 2013
8. Journal of Systems and Software (JSS), 2012

Research Grant Proposal Panelist/Reviewer

1. National Science Foundation (NSF), panelist, 2022
2. National Science Foundation (NSF), panelist, 2019, 2020
3. National Science Foundation (NSF), ad-hoc reviewer, 2020
4. Army Research Office (ARO), reviewer, 2019
5. FNR Luxembourg CORE programme, reviewer, 2018, 2019, 2020 (twice), 2022, 2023, 2024
6. Swiss National Science Foundation (SNSF), 2024, 2025
7. Research Grants Council (RGC) of Hong Kong, 2026

Graduate Student Committee Membership

1. Member of dissertation committee for Zheyuan Ma (UB Ph.D. student), 2026
2. Member of dissertation committee for Qiqing Huang (UB Ph.D. student), 2026
3. Member of dissertation committee for Yu Nong (my Ph.D. student at UB), 2026
4. Chair of oral qualifying-exam (OQE) committee for Jiawei Guo (my Ph.D. student at UB), 2025
5. Member of oral qualifying-exam (OQE) committee for Jue Guo (UB Ph.D. student), 2025
6. Chair of oral qualifying-exam (OQE) committee for Yu Nong (my Ph.D. student at UB), 2025
7. Member of oral qualifying-exam (OQE) committee for Qiqing Huang (UB Ph.D. student), 2025
8. Member of oral qualifying-exam (OQE) committee for Gaoxiang Liu (UB Ph.D. student), 2024
9. Chair of qualifying-exam (QE) committee for Tashi Stirewalt (WSU Ph.D. student), 2024

10. Chair of qualifying-exam (QE) committee for Mohammad Fakhruddin Babar (WSU Ph.D. student), 2023
11. Chair of qualifying-exam (QE) committee for Devjeet Raj Roy (WSU Ph.D. student), 2020
12. Chair of qualifying-exam (QE) committee for Ziyi Zhang (WSU Ph.D. student), 2020
13. Chair of qualifying-exam (QE) committee for Samir Sbair (WSU Ph.D. student), 2019
14. Chair of qualifying-exam (QE) committee for Neda Zarayeneh (WSU Ph.D. student), 2018
15. Chair of qualifying-exam (QE) committee for Zhengyang Wang (WSU Ph.D. student), 2018
16. Chair of qualifying-exam (QE) committee for Sarah Fakhoury (WSU Ph.D. student), 2018
17. Chair of qualifying-exam (QE) committee for Saghan Mudbhari (WSU Ph.D. student), 2017
18. Chair of qualifying-exam (QE) committee for Kudrat Jot Kaur (WSU Ph.D. student), 2017
19. Member of qualifying-exam (QE) committee for Funso Oje (WSU Ph.D. student), 2022
20. Member of qualifying-exam (QE) committee for Haoran Yang (my Ph.D. student at WSU), 2022
21. Member of qualifying-exam (QE) committee for Yu Nong (my Ph.D. student at WSU), 2021
22. Member of qualifying-exam (QE) committee for Wen Li (my Ph.D. student at WSU), 2020
23. Member of qualifying-exam (QE) committee for Iman Mirzadeh (WSU Ph.D. student), 2019
24. Member of qualifying-exam (QE) committee for Hanchao Ma (WSU Ph.D. student), 2019
25. Member of qualifying-exam (QE) committee for Xiaoqin Fu (my Ph.D. student at WSU), 2018
26. Member of qualifying-exam (QE) committee for Mohammad Omar Faruk (WSU Ph.D. student), 2018
27. Member of qualifying-exam (QE) committee for Md. Methun Kamrazzuman (WSU Ph.D. student), 2016
28. Chair of dissertation (and prelim-exam) committee for Yu Nong (my Ph.D. student at WSU), 2023
29. Chair of dissertation (and prelim-exam) committee for Wen Li (my Ph.D. student at WSU), graduated in 2023
30. Chair of dissertation (and prelim-exam) committee for Xiaoqin Fu (my Ph.D. student at WSU), graduated in 2022
31. Member of dissertation (and prelim-exam) committee for Ziyi Zhang (WSU Ph.D. student), 2023
32. Member of dissertation (and prelim-exam) committee for James Halvorsen (WSU Ph.D. student), 2022
33. Member of dissertation (and prelim-exam) committee for Neda Zarayeneh (WSU Ph.D. student), graduated in 2022
34. Member of dissertation (and prelim-exam) committee for Md. Methun Kamrazzuman (WSU Ph.D. student), graduated in 2021
35. Member of dissertation (and prelim-exam) committee for Md Rakibul Islam (WSU Ph.D. student), graduated in 2021
36. Co-chair of thesis committee for Shlok Tomar (Masters student at WSU, co-advised with Jana), 2023–2024
37. Co-chair of thesis committee for Dheeraj Vurukuti (Masters student at WSU, co-advised with Jana), 2023–2024
38. Chair of thesis committee for Pallavi Arivukkarasu (my Masters student at WSU), graduated in 2023
39. Chair of thesis committee for John Jenkins (my Masters student at WSU), graduated in 2020
40. Chair of thesis committee for Yu Nong (my Masters student at WSU), graduated in 2020
41. Member of thesis committee for Farooq Gessa (WSU Masters student), 2024
42. Member of thesis committee for James Laurence Crabb (WSU Masters student), 2022–2023

43. Member of thesis committee for Yan Zhao (Masters student at Virginia Tech), graduated in 2021
44. Member of thesis committee for Rui Zhou (Masters student at Concordia University, Canada), graduated in 2020
45. Member of thesis committee for Yuzhan Ma (WSU Masters student), graduated in 2017
46. Member of thesis committee for Ankita Tanwar (WSU Masters student), graduated in 2017
47. Member of thesis committee for Yongjun Chen (WSU Masters student), graduated in 2018
48. Member of thesis committee for Michael Christensen (WSU Masters student), graduated in 2020
49. Chair of advisory committee for Zhiping Li, Jingyang Ruan, and Jianqiao Liu (non-thesis masters students at WSU), graduated in 2022
50. Chair of advisory committee for Haoran Yang, Pengfei Ye, Mohammad Nouman Shafqat, and Nghia Nuong (non-thesis masters students at WSU), graduated in 2021
51. Chair of advisory committee for Weile Lian, Xin Da, Yashovardhan Sharma (all non-thesis masters students at WSU), graduated in 2020
52. Chair of advisory committee for Yuzhu Feng, Ziyi Zhang (all non-thesis masters students at WSU), graduated in 2019
53. Member of advisory committee for non-thesis masters students at WSU: Many including Afoke Hilda, Alfonso Ruelas, Garhgaj Singh, David Kelly, Afoke Abogidi, Scott Hawkins, Chris Nyugen, Deep Inamdar, Basu, Sreerupa; Njuguna, Jesse; Andrei kondyrev; Wu, Chiu-Yuan; Lee, Guang-Zheng; GeLinias, Philip; Gusa, Riley; Pilli, Eshwar; Bhavarisetty, Ramyasai; Inamdar, Deep; Xu, Tianxiang; Kondyrev, Andrei; Li, Wen-Chih; Mudassar, Zain; Shah, Dharmil Kamlesh; Deepesh Gupta; Medhi, Dipankar; Gade, Jaswanth Naidu; Ghanwat, Tejas; Lnu, Anshuman; Patil, Shruti Sunil; Doppalapudi, Nikhil Chakravathy; Lee Hungwei; Chao Jiang; Zhou Xiaonan; Lieh-Hsuan Chen; Kuang, Xiangtian; Wang, Yun-Ting; Lei Chen; Huang, Jingyuan; Lu, Jiayi; Luong, Nghia; Rainy Sharma; Alexander Joens; Njuguna, Jesse; Daniel Olds; Ran Chaowen; Junhao Wang; Munir, Tayyab; Han, Namgyu; Nimbekar, Pradnya

Intramural Services at UB

1. Member of faculty search committee (computer security), Spring 2025 -
2. Internship Coordinator, Spring 2025 -
3. Member of Graduate Admission Committee, Fall 2024 - Spring 2025

Intramural Services at WSU

1. Coordinator of Software Engineering program assessment for ABET, Fall 2023 -
2. Member of EECS ABET committee, Fall 2023 -
3. Chair of career track faculty search committee, Fall 2023 - Spring 2024
4. Chair of junior faculty mentoring committee (Monowar Hasan), Spring 2023 - Spring 2024
5. Tenure & Promotion committee member (for Feng-Hao Liu), Spring 2023
6. Member of Graduate Study Committee (GSC) assistant search committee, Spring 2023
7. Member of EECS tenured/tenure-track faculty search committee (Security area, Pullman campus), Summer 2022 – Spring 2023
8. Member of EECS tenured/tenure-track faculty search committee (AI/ML area, Pullman campus), Summer 2022 – Spring 2023
9. Member of VCEA Cybersecurity faculty search committee (Everett campus), Summer 2022 – Spring 2023
10. Member of VCEA Cybersecurity curriculum development committee, Fall 2022 – Spring 2023
11. Chair of VCEA Cybersecurity research committee, Fall 2022 – Spring 2023
12. Member of VCEA Cybersecurity coordinating committee, Fall 2022 – Spring 2023

13. Member of EECS tenured/tenure-track faculty search committee (AI/ML/Security/Systems areas), Fall 2021 – Spring 2022
14. Member of *Experience WSU* (an annual event for undergraduate student recruitment) faculty team, Spring 2018, 2019, 2024
15. Member of EECS career-track faculty search committee, Spring 2022 – Summer 2022
16. Member of EECS Computer Science Curriculum Committee (CSCC), since Fall 2021
17. Member of EECS faculty search committee (systems area), Fall 2018 – Spring 2019
18. Member of EECS Graduate Studies Committee (GSC), since Fall 2018
19. Course coordinator for CptS 428/528, since Fall 2022
20. Course coordinator for CptS 422, since Fall 2018
21. Course coordinator for CptS 322, since Fall 2016

Other Services

1. Mentor, Individualized Cybersecurity Research Mentoring (iMentor) Workshop (collocated with ACM CCS), 2023
2. Reviewer/Judge, Undergraduate student research exhibition (Academic Showcase), Pullman, 2023
3. Reviewer/Judge, DARPA Riser Event at DARPA Forward Conference, Pullman, 2022
4. Student Volunteer, IEEE VisWeek 2011
5. Interviewer, Technical Recruitment Team of the Web-Search Dept., Baidu Inc. 2009, 2010

PROFESSIONAL SOCIETY MEMBERSHIPS

1. The Advanced Computing Systems Association (USENIX), Champion member 2021 - present
2. The Institute of Electrical and Electronics Engineers (IEEE), **Senior member** 2023 - present
3. The Association for Computing Machinery (ACM), **Senior member** 2023 - present
4. The Institute of Electrical and Electronics Engineers (IEEE), Member 2014 - 2022
5. The Association for Computing Machinery (ACM), Member 2016 - 2022
6. ACM Special Interest Group on Software Engineering (SIGSOFT) 2016 - present

INVITED SEMINAR TALKS

1. Learning-Based Code Analysis and Security/Privacy Defenses for Mobile Applications
Keynote speech at the 13th International Conference on Mobile Software Engineering and Systems (MOBILESoft) 2026, Co-located with ICSE 2026, Rio de Janeiro, Brazil April, 2026
2. Cost-Effective Program Analysis for Secure Distributed Software Systems
Invited talk at ESL Global Cybersecurity Institute, Rochester Institute of Technology, Rochester, NY April 2026
3. Sustainable Defense against Mobile Malware
Keynote speech at the 2030 Software Engineering, Co-located with FSE 2025, Trondheim, Norway June, 2025
4. The Data Quest in Software Vulnerability Analysis: Hope, Challenges, and New Horizon
Keynote speech at the 21st International Conference on Predictive Models and Data Analytics in Software Engineering (PROMISE), Co-located with FSE 2025, Trondheim, Norway June, 2025
5. Automated Generation of Quality Vulnerability Data at Scale
Keynote speech at The International Workshop on Software Vulnerability Management (SVM), Co-located with ICSE 2025, Ottawa, Canada May, 2025 (invited)
6. Program Analysis and Security for Enterprise-Scale Distributed Software Systems
Upbeats Talk at CSE Department, University at Buffalo, Buffalo, NY October, 2024
7. Empowering learning-based vulnerability analysis via automated data augmentation
Northwest Virtual Institute for Cybersecurity Education and Research (CySER) workshop, Pullman, WA May 2024

8. Sustainable defenses against evolving mobile malware
Northwest Virtual Institute for Cybersecurity Education and Research (CySER) workshop,
Pullman, WA May 2023
9. Cost-effective analysis and security defenses for enterprise-level distributed systems
IEEE Palouse Section research seminar, Pullman, WA January 2023
10. Defending the information flow security of enterprise-scale distributed systems
Northwest Virtual Institute for Cybersecurity Education and Research (CySER) seminars,
Pullman, WA December 2022
11. Smart phone technology security: the case of Android: systems, attacks, defenses
Northwest Virtual Institute for Cybersecurity Education and Research (CySER) workshop,
Pullman, WA June 2022
12. Cost-effective and data-driven program analysis for the reliability and security of real-world
software systems
WSU EECS Advisory Board meeting, Pullman, WA April 2022
13. Integrated program analysis for software reliability and security
WSU EECS Advisory Board meeting, Pullman, WA September 2018
14. Program analyses for the reliability and security of evolving software
WSU EECS CptS500 (seminar course), Pullman, WA November 2016
15. Advanced hybrid program analyses for software reliability and security
WSU EECS Advisory Board meeting, Boise, ID October 2016
16. Cost-effective program dependence analyses for evolving and securing modern software
School of EECS, Washington State University, WA April 2016
17. Cost-effective program dependence analyses for reliable and secure software evolution
CS Department, Missouri University of Science and Technology, MO March 2016
18. Cost-effective program dependence analyses for evolving and securing modern software
CS Department, University of Massachusetts, Boston, MA March 2016
19. A hybrid approach to dependence approximation for dynamic impact prediction
ECE Department, Virginia Tech, VA October 2015
20. Semantic dependence quantification using sensitive analysis
CS Department, Virginia Tech, VA September 2015
21. Cost-effective dependence analysis for dynamic impact prediction
CS Department, University of Illinois, Urbana Champaign, IL July 2015
22. Sensitivity analysis for quantitative change-impact prediction
GrammaTech, Inc., NY June 2015

CONFERENCE PANELS & TALKS

1. Two-Level Adaptation for Budget-Constrained Continuous Dynamic Dependence Analysis.
ACM International Conference on the Foundations of Software Engineering (FSE), Montreal,
Canada July 2026
2. A Practical Fuzzer for the Python Runtime System.
ACM International Conference on the Foundations of Software Engineering (FSE), Montreal,
Canada July 2026
3. How Are Multilingual Systems Constructed: Characterizing Language Use and Selection in
Open-Source Multilingual Software.
IEEE/ACM International Conference on Software Engineering (ICSE), Lisbon, Portugal April
2024
4. PyRTFuzz: Detecting Bugs in Python Runtimes via Two-Level Collaborative Fuzzing.
ACM Conference on Computer and Communications Security (CCS), Copenhagen, Denmark
November 2023
5. Fuzzing I (Approach) (Paper session chair).
ACM Conference on Computer and Communications Security (CCS), Copenhagen, Denmark
November 2023

6. VulGen: Realistic Vulnerable Sample Generation via Pattern Mining and Deep Learning.
IEEE/ACM International Conference on Software Engineering (ICSE), Melbourne, Australia
May 2023
7. Demystifying Issues, Challenges, and Solutions for Multilingual Software Development.
IEEE/ACM International Conference on Software Engineering (ICSE), Melbourne, Australia
May 2023
8. Open Science in Software Engineering: A study on Deep Learning-based Vulnerability Detection.
IEEE/ACM International Conference on Software Engineering (ICSE), Melbourne, Australia
May 2023
9. Empirical Evaluation of Secure Development Processes (Panelist).
Dagstuhl Seminar 23181, Dagstuhl, Germany May 2023
10. Towards Unified and Sustainable Dynamic Malware Forensics.
International Conference on Network and Information Security, Hangzhou December 2022
11. Generating Realistic Vulnerabilities via Neural Code Editing: An Empirical Study.
ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Singapore November 2022
12. Language-Agnostic Dynamic Analysis of Multilingual Code: Promises, Pitfalls, and Prospects.
ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Singapore November 2022
13. Program Analysis - FSE'20 papers (Paper session chair).
ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Singapore November 2022
14. Program Analysis - FSE'21 papers (Paper session chair).
ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Singapore November 2022
15. Mobile Health & Covid-19 Apps (Paper session chair).
IEEE/ACM International Conference on Mobile Software Engineering and Systems (MOBILESoft), Virtual May 2022
16. Code Summarization (Paper session chair).
IEEE/ACM International Conference on Program Comprehension (ICPC), Virtual May 2022
17. Mobile Software Engineering (Panelist).
ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Virtual November 2020
18. Software Analysis (Panelist).
ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Virtual November 2020
19. Towards Learning Visual Semantics.
ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Virtual November 2020
20. Assessing and Improving Malware Detection Sustainability through App Evolution Studies.
ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Virtual November 2020
21. A Study of Run-time Behavioral Evolution of Benign versus Malicious Apps in Android.
International Conference on Software Maintenance and Evolution (ICSME), Virtual July 2020
22. Embracing Mobile App Evolution via Continuous Ecosystem Mining and Characterization.
International Conference on Mobile Software Engineering and Systems (MOBILESoft), Virtual July 2020
23. A Reflection on the Predictive Accuracy of Dynamic Impact Analysis.
International Conference on Software Analysis, Evolution, and Reengineering (SANER), Vir-

tual
2020

February

24. Applied Mathematics, Computer Science, and Data Science Foundations for U.S. Department of Energy Mission Cyber Resiliency (Panel).
PNNL CENATE Cybersecurity Roundtable Event, Richland, WA August 2019
25. A large-scale study of application incompatibilities in Android.
ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), Beijing, China July 2019
26. Understanding android application programming and security: A dynamic study.
International Conference on Software Maintenance and Evolution (ICSME), Shanghai, China September 2017
27. Droidfax: A toolkit for systematic characterization of android applications.
International Conference on Software Maintenance and Evolution (ICSME), Shanghai, China September 2017
28. Artifacts for dynamic analysis of android apps.
International Conference on Software Maintenance and Evolution (ICSME), Shanghai, China September 2017
29. Dissecting android inter-component communications via interactive visual explorations.
International Conference on Software Maintenance and Evolution (ICSME), Shanghai, China September 2017
30. DistIA: a cost-effective dynamic impact analysis for distributed programs
International Conference on Automated Software Engineering (ASE), Singapore September 2016
31. Abstracting program dependencies using the method dependence graph
International Conference on Software Quality, Reliability, and Security (QRS), Vancouver, BC August 2015
32. Facilitating information management in integrated development environments through visual interface enhancements
International Conference on Software Quality, Reliability, and Security (QRS), Vancouver, BC August 2015
33. TracerJD: generic trace-based dynamic dependence analysis with fine-grained logging
International Conference on Software Analysis, Evolution, and Reengineering (SANER), Montreal, QB March 2015
34. A framework for cost-effective dependence-based dynamic impact analysis
International Conference on Software Analysis, Evolution, and Reengineering (SANER), Montreal, QB March 2015
35. SensA: sensitivity analysis for quantitative change-impact prediction
International Conference on Source Code Analysis and Manipulation (SCAM), Victoria, BC September 2014
36. Estimating the accuracy of dynamic change-impact analysis using sensitivity analysis
International Conference on Software Security and Reliability (SERE), San Francisco, CA June 2014

TECHNICAL SKILLS

Programming Languages professional development in C/C++ (400 KSLOC), Python (90 KSLOC), Java (100 KSLOC), Shell/Awk (50 KSLOC), and Delphi (20 KSLOC); versed in Scala, SQL, HTML, Matlab and R; experienced in SAS, JavaScript, ActionScript, PHP and XML

SDK/Frameworks skilled programming using the Soot Java optimization framework (80 KSLOC), OpenGL&GLUT (80 KSLOC), VTK (30 KSLOC); experienced in Android SDK, openMPI, Qt, MFC, and Xlib/X11

Development Environments proficient in Linux & Windows application and system/network software development, with proficiency in event-driven, parallel, and asynchronous programming paradigms

**SELECTED
CERTIFICATES**

1. *Technical Project Management*, American Management Association (AMA), 2010
2. *Certified System Analyst*, Ministry of Information Industry of China, (pass rate 5%), 2007
3. *Certified Software Engineer*, Ministry of Information Industry of China (pass rate 15%), 2003